

ToolUniverse

Give any AI model a standardized toolbox of vetted scientific instruments

CCB AI SEMINAR & WORKSHOP SERIES · HARVARD MEDICAL SCHOOL

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What you will be able to do

- | | | |
|----|---|---------------|
| 01 | Explain why a chatbot alone is not enough for scientific work | Reza |
| 02 | Install ToolUniverse on your own machine | Yepeng |
| 03 | Recognise what ToolUniverse is, and what it is not | Reza |
| 04 | Run a query using ToolUniverse live | Reza & Yepeng |
-

Who has been handed a citation
that turned out not to exist?

It got past expert reviewers at NeurIPS 2025

100

fabricated citations
confirmed

53

accepted papers, each
with 3+ expert reviewers

66%

total fabrications

The same process invents facts, not just citations

TP53 interacts with ? ← the most *plausible* next word

A citation

Easy to catch: you go look it up

A protein interaction

Hard to catch: looks real and you may trust it

A dose

Hard to catch: a wrong number that looks right

First, let's get it installed

What you need

- 01 A laptop and your preferred AI assistant (ChatGPT app or Claude Desktop recommended).
- 02 Internet for the first launch (about 160 Python packages).
- 03 (Optional) API keys for certain resources (e.g., BioGRID, NCBI, OMIM, etc.).

	ANTHROPIC	OPENAI
Desktop	Claude Desktop	ChatGPT app (Work)
CLI	Claude Code	Codex CLI
Underlying LLM	Claude	GPT

Components in ToolUniverse are identical once it is in.

The whole install is one sentence

PASTE INTO CLAUDE DESKTOP (COWORK), CLAUDE CODE, CHATGPT APP, OR CODEX CLI

“Read <https://aiscientist.tools/setup.md> and set up ToolUniverse for me.”

-
- 01 The assistant reads the official setup guide, asks whether you want chat, command line, or Python, installs the one prerequisite (uv), installs ToolUniverse, edits its own config file, and runs a test query.
 - 02 The installation process takes about ten minutes.
 - 03 If anything goes wrong, paste the error back into the same chat.
-

What that sentence did (if you want to do it by hand)

Prerequisite: install uv (a Python package manager) with:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
# Windows: astral.sh/uv/install.ps1
```

Then one line registers the same toolbox with your client:

```
ANTHROPIC      # Claude Code
                claude mcp add -s user tooluniverse -e PYTHONIOENCODING=utf-8 -- uvx tooluniverse
                # Claude Desktop: Settings > Developer > Edit Config, add a "tooluniverse" entry

OPENAI         # Codex CLI and Codex app (shared config)
                codex mcp add tooluniverse --env PYTHONIOENCODING=utf-8 -- uvx tooluniverse
                # writes [mcp_servers.tooluniverse] to ~/.codex/config.toml
```

Claude Desktop also has a zero-terminal route: **Settings → Extensions → Browse → ToolUniverse → Install.**

Verify: is it actually connected?

VERIFICATION PROMPT, ANY CLIENT

“Using ToolUniverse, tell me how many tools are loaded and name five tool categories. Then run PubMed_search_articles with query ‘CRISPR’ and max_results 1, and show me the PMID.”

You should see

A tool call to `find_tools` or `list_tools`

Then `execute_tool`, then a real PMID and title

About 2,700 tools loaded

Normal, not broken

First call after install: 30–60 s while packages load

Later calls: seconds

When it goes wrong: the four fixes that cover most cases

01 **uvx: command not found**

uv installed but the terminal was not reopened. Reopen it. In a desktop app's config, use the *absolute* path to uvx; GUI apps do not see your shell's PATH.

02 **The app shows no tools**

It was closed, not quit. Cmd+Q (macOS) or quit from the tray (Windows), then relaunch. Claude Desktop: check the JSON. Codex: check ~/.codex/config.toml.

03 **The first call hangs**

It is downloading. Wait a minute. On slow Wi-Fi, warm the cache once in a terminal:
`uvx tooluniverse --help.`

04 **externally-managed-environment**

You used system pip.
Use `uv pip install tooluniverse` instead.

Or paste the error into the same chat that did the install and say “fix it.” tooluniverse-doctor will print what is missing and why.

“What interacts with TP53?”

CHATBOT, NO TOOLS

Based on my training data, TP53 interacts with MDM2, BRCA1, and
ATM...

[MAY BE OUTDATED]

No source No date No confidence

What web search still can't do

Ask your own question

Your compound, your target, your cut-off: no page matches that exactly

Run an analysis

e.g. which pathways your gene list hits or how tightly a new molecule binds

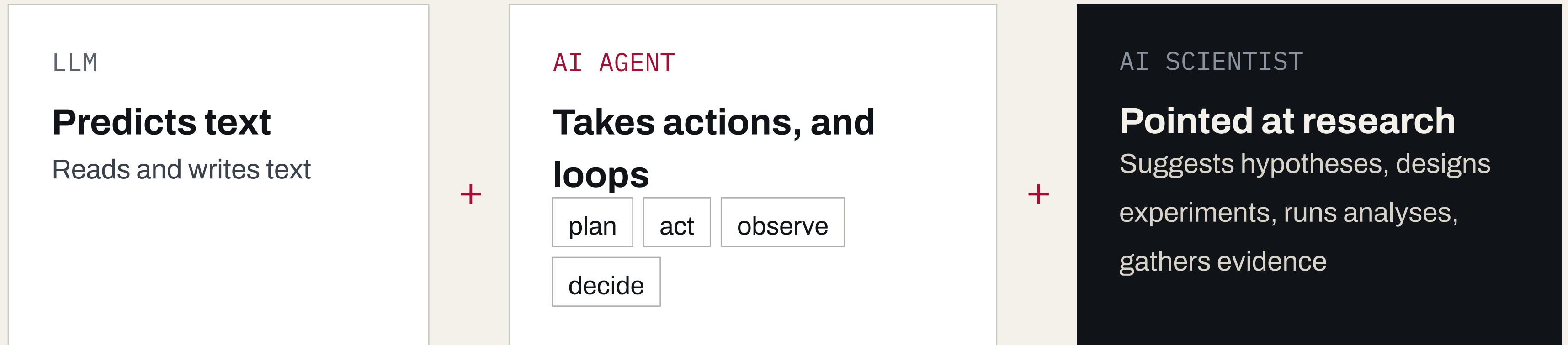
Show where it came from

A logged database call can go in your methods and be rerun

Hand back exact values

A database returns the number itself; a web page has to be read and rewritten by the AI

From chatbot to AI scientist



Computational *co*-scientists: you stay in the loop.

Definitions: Tool, API, MCP

Tool

Anything the AI can use to
get a real result

*e.g. a STRING lookup, a
binding model, a PubMed
search*

API (Application Programming Interface)

A tool's own way of taking
requests

STRING's and PubMed's
look nothing alike

MCP (Model Context Protocol)

One shared plug between
the AI and all its tools

So the AI never has to learn
each API

MCP is the plug. ToolUniverse is the translator behind it, for 2,700+ tools.

“What interacts with TP53?”

WITH TOOLUNIVERSE

249 experimentally validated partners from Reactome/IntAct.

Top: MDM2 (score 0.995, 122 evidences), EP300, MDM4,

USP7...

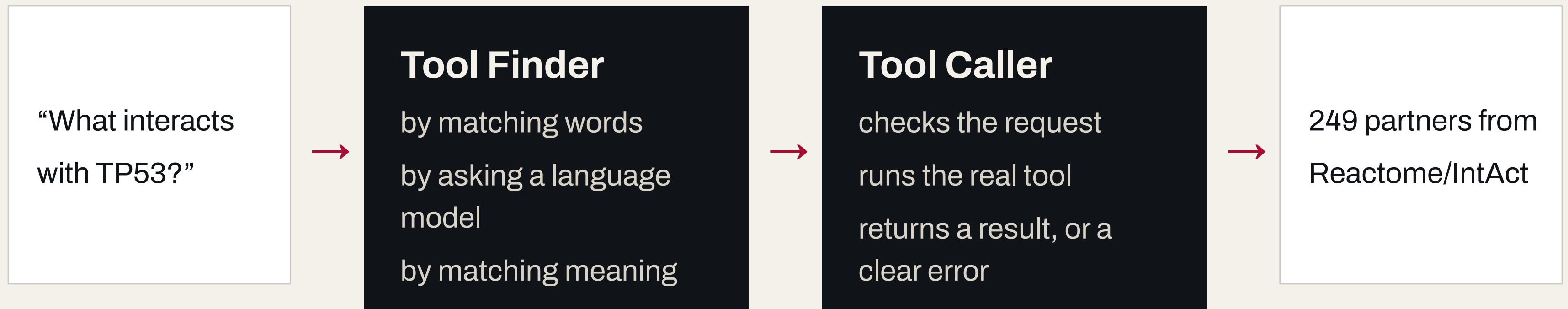
[LIVE DATA, SOURCED]

IntAct is a curated database of interactions from published experiments. Each pair gets a confidence score from 0 to 1, based on how much evidence supports it, and of what kind.

ToolUniverse

An open, free platform that gives any AI model a standardized toolbox of vetted scientific instruments, plus the machinery to find the right one, use it correctly, chain several together, and build new ones.

Find the right tool, then call it



Add your own tools, then chain them

Tool Manager

Local: a short spec plus a little code

Remote: private data and GPU models join,
nothing exposed

Tool Composer

search + search + search → summarize

In sequence, in parallel, or in a loop

Build new tools, and keep them honest

Tool Discoverer



Tool Optimizer

Run the tool on test cases

Compare its description to what it does

Rewrite the description

The whole picture



Bring your own model. Switch any time, no retraining.

Every tool describes itself the same way

01

**What it's for, in plain
language**

So the AI can pick the right one

02

**Exactly what goes in
and comes out**

So bad requests fail loudly

03

One way to call it

Whatever is running underneath

What is in the library

2,700+
scientific tools

180+
research skills



Databases

OpenTargets · ChEMBL · UniProt · openFDA · ClinicalTrials.gov

ML models

Boltz-2 · ADMET-AI · SpliceAI

Literature

PubMed · Europe PMC · Semantic Scholar · bioRxiv

Agents and workflows

Hypothesis generation · literature review · biomarker discovery

Fields it covers

Genomics

Transcriptomics

Proteomics

Metabolomics

Multi-omics

Immunology

Pharmacology

Small-molecule
chemistry

Clinical medicine

GWAS & genetics

Who built it

Zitnik Lab · Department of Biomedical Informatics, HMS

With Harvard, MIT, MIT Lincoln Laboratory, the Broad Institute, and the Kempner Institute.

Lead creator Shanghua Gao · PI Marinka Zitnik

Model-agnostic

Any model, no retraining

Standardized

One format for every tool

Open source

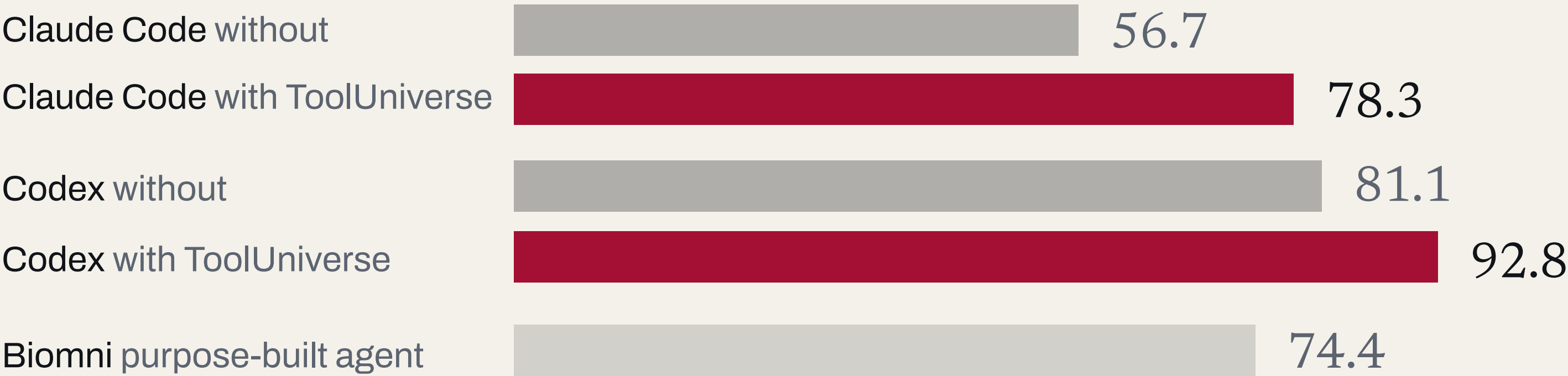
Apache 2.0, free

Checked

Per-tool tests + human verified

Does it hold up?

Same agent, same model, **with vs. without ToolUniverse** · LAB-Bench database questions (DbQA), accuracy %



Sequence questions (SeqQA): Claude Code 94.3 → 99.5 · Codex 91.4 → 96.2

EARLIER RESULT · ORIGINAL 211-TOOL VERSION

TxAgent 92.1% vs GPT-4o 66.3% on drug reasoning

IN USE

500,000+ agent analyses (May 2026)

1.1M+ downloads from 131 countries (Sep 2026)

What to check before you trust it

-
- 01 Predictions still need experimental validation
-
- 02 Correct data can still lead to a wrong conclusion.
Read the reasoning, not just the answer.
-

Sources

Gao et al., ToolUniverse: An open platform for democratizing AI scientists

Nature Methods (in press) · [arXiv:2509.23426](#)

ToolUniverse platform and code

[aiscientist.tools](#) · [github.com/mims-harvard/ToolUniverse](#)

Protein interaction databases

[reactome.org](#) · [ebi.ac.uk/intact](#) · [string-db.org](#)

Gao et al., TxAgent

[arXiv:2503.10970](#)

Fabricated citations at NeurIPS 2025

GPTZero scan, Jan 2026 · Ansari, [arXiv:2602.05930](#)

Usage and downloads

Zitnik Lab, May 2026 · PyPI via ClickPy, Sep 2026

Now, let's run it

In Claude

“Using ToolUniverse, what interacts with TP53?”

Watch what it does

Which tools it picks, which database it uses, what comes back

Then, two worked examples

Clinical and genetics step by step

Two example questions

For each question: the reasoning Claude wrote out, every ToolUniverse call, and what came back. All calls were run live on October 5th, 2026.

01

Apixaban or warfarin at an eGFR of 22, and does the evidence cover this patient?

CLINICAL

PubMed_search_articles → PubMed_search_articles → PubMed_get_article_metadata → PubMed_search_articles

First search: 27,459 hits, because PubMed misread the query. Second: the wrong patients. Claude caught both.

02

Which gene does a non-coding variant control, and in which tissue?

GENETICS

dbSNP_get_variant_by_rsid → GTEx_get_single_tissue_eqtls → UCSC_get_tf_binding_clusters → PubMed_search_articles

The association in ToolUniverse return is the wrong answer. The tissue has to come from the phenotype.

A clinical question

THE QUESTION

“A 66-year-old has atrial fibrillation and an eGFR of 22, which is chronic kidney disease stage 4 and not dialysis. Apixaban or warfarin? The pivotal trials excluded this degree of renal impairment, so the real question is whether the retrievable literature is about this patient at all.”

WHAT EVERY TERM IN IT MEANS

atrial fibrillation	An irregular heart rhythm that lets clots form in the atrium. The reason to anticoagulate at all is to prevent a stroke.
eGFR 22	Estimated glomerular filtration rate, in mL/min/1.73m ² . Normal is above 90. At 22 the kidneys clear drugs slowly, and apixaban is partly renally cleared.
CKD stage 4	Chronic kidney disease staged by eGFR. Stage 4 is 15 to 29, severe but pre-dialysis. Stage 5 and dialysis are a different population with different evidence.
apixaban	A direct oral factor Xa inhibitor, dosed on a rule written in serum creatinine rather than eGFR.
warfarin	A vitamin K antagonist, not renally cleared, but needing INR monitoring and unstable in kidney disease.
pivotal trials	ARISTOTLE and its contemporaries, which licensed these drugs. They enrolled almost nobody at this level of renal function.

THE CALLS, AND WHAT EACH ONE IS FOR

- 01 **PubMed_search_articles**
Ask the way a clinician would phrase it, then read what PubMed did to the query.
- 02 **PubMed_search_articles**
Ask again with field tags, and inspect the population the new query selected.
- 03 **PubMed_get_article_metadata**
Read what each retrieved study is about, rather than ranking them.
- 04 **PubMed_search_articles**
Correct the population and search once more, excluding dialysis.

Claude and ToolUniverse, turn by turn

claude > thinking

CLINICAL · CALL 1 OF 4

I will search the way the question was asked, and check what PubMed makes of the query before I trust a count.

claude > tool_use PubMed_search_articles

```
{"query": "apixaban warfarin atrial fibrillation stage 4 OR stage 5 chronic kidney disease", "limit": 10}
```

tooluniverse > result count 27459 · retmax 10 · retstart 0

```
idlist 39680399, 33720773, 33313680, 23486521, 30144250, 33538928, 34270930, 41085503, 37960159, 29105079
```

querytranslation

```
((("apixaban"[Supplementary Concept] OR "apixaban"[All Fields] OR "apixaban s"[All Fields])
AND ("warfarin"[Supplementary Concept] OR "warfarin"[MeSH Terms] OR "warfarinization"[All Fields] ...)
AND ("atrial fibrillation"[MeSH Terms] OR ("atrial"[All Fields] AND "fibrillation"[All Fields]) ...)
AND ("stage"[All Fields] OR "staged"[All Fields] OR "stages"[All Fields] ...) AND "4"[All Fields])
```

OR

```
((("stage"[All Fields] OR "staged"[All Fields] OR "stages"[All Fields] ...) AND "5"[All Fields]
AND ("renal insufficiency, chronic"[MeSH Terms] OR ("chronic"[All Fields] AND "kidney"[All Fields]
AND "disease"[All Fields]) OR "chronic kidney disease"[All Fields]))
```

claude > thinking

27,459 is not a literature. PubMed split the query at my OR, and the second half of the translation contains neither apixaban nor warfarin: it is every paper that mentions a stage, the digit 5, and chronic kidney disease. The count looks like a large and reassuring evidence base and is an artefact of query parsing. I will ask again with explicit field tags so that nothing is rewritten behind me.

Claude and ToolUniverse, turn by turn

```

claude > tool_use PubMed_search_articles
{"query": "apixaban[tiab] AND warfarin[tiab] AND \"atrial fibrillation\"[MeSH]
        AND (\"kidney failure, chronic\"[MeSH] OR \"renal dialysis\"[MeSH])", "limit": 10}

tooluniverse > result count 35 • translationset empty, nothing was rewritten
idlist 36335914, 31976865, 29954737, 30354526, 41085503, 31209952, 27079417, 36469596, 23790601, 30176675

claude > tool_use PubMed_get_article_metadata
{"pmids": ["36335914", "31976865", "29954737", "30354526", "41085503", "31209952"]}

tooluniverse > result

| PMID     | YEAR | JOURNAL                | TYPE                  | TITLE                                                     |
|----------|------|------------------------|-----------------------|-----------------------------------------------------------|
| 36335914 | 2022 | Circulation            | Randomised trial      | Apixaban for Patients With AF on Hemodialysis             |
| 31976865 | 2020 | J Am Coll Cardiol      | Network meta-analysis | Oral Anticoagulation for AF on Long-Term Hemodialysis     |
| 29954737 | 2018 | Circulation            | Observational         | Apixaban Use in End-Stage Kidney Disease and AF           |
| 41085503 | 2025 | JACC Cardiovasc Interv | Observational         | Stroke Prevention in AF and End-Stage Renal Disease       |
| 30354526 | 2018 | Circulation            | Editorial             | Long-Term Anticoagulation for Patients Receiving Dialysis |
| 31209952 | 2019 | Semin Dial             | Review                | Warfarin in nonvalvular AF, time for a change?            |

claude > thinking

Thirty-five clean hits, and every one of them is haemodialysis or end-stage kidney disease. That is my own fault: the MeSH terms I chose, kidney failure, chronic and renal dialysis, select the dialysis population. This patient has an eGFR of 22, which is stage 4 and not on dialysis. I have retrieved a neighbouring patient. The one randomised trial here, RENAL-AF, enrolled dialysis patients. I need to search again and exclude them.
```

Claude and ToolUniverse, turn by turn

CLINICAL · CALL 4 OF 4

claude > tool_use PubMed_search_articles

{ "query": "apixaban[tiab] AND warfarin[tiab] AND \"atrial fibrillation\"[MeSH] AND \"renal insufficiency, chronic\"[MeSH] NOT \"renal dialysis\"[MeSH]", "limit": 10 }

tooluniverse > result count 49 · translationset empty

idlist	PMID	YEAR	JOURNAL	TYPE	TITLE
38035566, 36315950, 37839687, 33640261, 41085503, 32740218, 36469596, 34459281, 29105079, 22825670					
	37839687	2024	Am J Kidney Dis	Nationwide cohort	Warfarin or Rivaroxaban vs Apixaban in Advanced CKD and AF
	38035566	2024	Am J Nephrol	Review	Anticoagulation in Patients with Chronic Kidney Disease
	36315950	2022	Ann Intern Med	Multinational cohort	Comparative Effectiveness and Safety Between Four DOACs in AF
	32740218	2020	Curr Opin Nephrol Hy	Review	Direct oral anticoagulants in chronic kidney disease, an update
	33640261	2021	J Stroke Cerebrovasc	Review	Controversies: Stroke Prevention in Chronic Kidney Disease
	34459281	2022	Ann Pharmacother	Review	DOACs in Patients With Stage 5 or End-Stage Kidney Disease

claude > thinking

PMID 37839687 is the first record in four calls that is actually about this patient: a nationwide US cohort in advanced chronic kidney disease, not on dialysis, comparing warfarin and rivaroxaban against apixaban. It is observational. There is still no randomised trial in this population, and the rest of the list is reviews.

claude > answer

No randomised evidence exists for a 66-year-old at eGFR 22. The best available comparison is Fu and colleagues, Am J Kidney Dis 2024, PMID 37839687, a nationwide cohort in advanced CKD without dialysis. That is what a clinician should be handed, with its design named, rather than a recommendation extrapolated from the dialysis trials that the first two searches returned.

A genetics question

THE QUESTION

“rs12740374 is a common non-coding variant at the 1p13 locus, the strongest genome-wide association signal for LDL cholesterol and myocardial infarction. It sits inside the 3’ untranslated region of CELSR2, but it changes no protein. Which gene does it actually regulate, in which tissue?”

WHAT EVERY TERM IN IT MEANS

rs12740374	A reference SNP identifier. One specific single-base change, catalogued in dbSNP, here a G to T substitution on chromosome 1.
non-coding	The variant lies outside any protein-coding sequence, so it cannot change a protein. If it does anything, it changes how much of some gene is made.
1p13 locus	The region of chromosome 1 where the signal sits. Six genes lie within a hundred kilobases, and three are the classic cluster: CELSR2, PSRC1, SORT1.
LDL cholesterol	Low-density lipoprotein cholesterol, cleared from the blood by the liver. Which is why the liver is the tissue this question has to be answered in.
eQTL	Expression quantitative trait locus. An association between carrying one allele and making more or less of a gene, measured per tissue.
3’ UTR of CELSR2	The untranslated tail of the CELSR2 transcript. dbSNP files the variant under CELSR2 for this reason, which is a fact about coordinates.

THE CALLS, AND WHAT EACH ONE IS FOR

- 01 dbSNP_get_variant_by_rsid**
Establish what the database claims, so there is something to overturn.
- 02 GTEx_get_single_tissue_eqtl**
Ask which genes the variant is associated with, across every tissue at once.
- 03 UCSC_get_tf_binding_clusters**
Ask whether the locus is a regulatory element at all, and which factors occupy it.
- 04 PubMed_search_articles**
Ask the literature whether anyone perturbed the variant itself.

Claude and ToolUniverse, turn by turn

claude > thinking

The variant is non-coding, so whatever gene dbSNP names will simply be the gene it sits inside. I need that answer first, as the thing to test.

claude > tool_use dbSNP_get_variant_by_rsid

`{"rsid": "rs12740374"}`

tooluniverse > result

```
"status": "success",
"data": {
  "refsnp_id": "rs12740374",
  "snps_id": 12740374,
  "chromosome": "1",
  "position": "1:109274968",
  "assembly": "GRCh38",
  "position_grch37": "1:109817590",
  "allele": "K",
  "snps_class": "snv",
  "clinical_significance": ["association"],
  "genes": ["CELSR2"],
  "allele_frequencies": [
    {"study": "1000Genomes", "freq": "T=0.195487/979"},
    {"study": "TOPMED", "freq": "T=0.222404/58868"},
    {"study": "ALFA", "freq": "T=0.210673/42223"}],
  "hgvs_notation": "NC_000001.11:g.109274968G>T,
    NM_001408.3:c.*919G>T|GENE=CELSR2:1952",
  "spdi_notation": "NC_000001.11:109274967:G:T",
  "function_class": ["3_prime_UTR_variant"],
  "validated": ["by-frequency", "by-alfa", "by-cluster"],
  "created_date": "2004/04/05 11:05",
  "updated_date": "2024/11/02 17:37"}
```

claude > thinking

CELSR2, on no evidence beyond position, and the only clinical note is "association". Expression data can tell me which gene the allele actually moves. I will ask every tissue at once, so the tissue becomes something I have to justify.

GENETICS · CALL 1 OF 4

Claude and ToolUniverse, turn by turn

claude > tool_use GTEx_get_single_tissue_eqtls

GENETICS · CALL 2 OF 4

{"variant_id": ["chr1_109274968_G_T_b38"], "dataset_id": "gtex_v8"}

tooluniverse > result 39 tissue-gene pairs, complete											
TISSUE	GENE	p	nes	TISSUE	GENE	p	nes	TISSUE	GENE	p	nes
Muscle_Skeletal	CELSR2	2.30e-76	+0.54	Esophagus_Mucosa	SORT1	8.42e-11	+0.13	Adipose_Subcut	PSRC1	1.38e-06	+0.20
Liver	SORT1	3.09e-54	+1.27	Nerve_Tibial	PSRC1	9.66e-09	+0.16	Adipose_Visceral	PSRC1	3.49e-06	+0.22
Liver	PSRC1	1.05e-37	+1.24	Minor_Salivary	PSRC1	4.46e-09	+0.51	Minor_Salivary	SORT1	7.30e-06	+0.36
Liver	CELSR2	9.43e-34	+1.04	Brain_Cortex	PSRC1	2.51e-09	+0.34	Stomach	PSRC1	7.17e-06	+0.19
Esophagus_Mucosa	PSRC1	1.81e-24	+0.40	Cells_Fibroblast	PSRC1	1.62e-08	+0.11	Skin_Sun_Exposed	SORT1	9.15e-06	+0.12
Esophagus_Mucosa	CELSR2	1.08e-16	+0.31	Pituitary	PSRC1	6.19e-08	+0.50	Skin_Not_Sun_Exp	CELSR2	6.10e-06	+0.10
Whole_Blood	PSRC1	4.03e-16	+0.30	Minor_Salivary	CELSR2	1.36e-07	+0.34	Colon_Sigmoid	PSRC1	1.42e-05	+0.27
Pancreas	CELSR2	2.82e-13	+0.32	Skin_Sun_Exposed	PSRC1	2.29e-07	+0.21	Prostate	PSRC1	1.87e-05	+0.31
Pancreas	PSRC1	2.22e-13	+0.53	Breast_Mammary	PSRC1	3.61e-07	+0.23	Skin_Sun_Exposed	GSTM4	2.29e-05	+0.23
Muscle_Skeletal	PSRC1	1.49e-12	+0.35	Heart_Left_Vent	PSRC1	4.51e-07	+0.26	Brain_Putamen	PSRC1	3.02e-05	+0.22
Skin_Not_Sun_Exp	PSRC1	2.03e-12	+0.29	Liver	ATXN7L2	5.86e-07	+0.40	Lung	GSTM2	7.39e-05	-0.13
Testis	PSRC1	9.00e-12	+0.35	Thyroid	PSRC1	8.84e-07	+0.20	Skin_Not_Sun_Exp	GSTM4	9.36e-05	+0.23
Skin_Sun_Exposed	CELSR2	5.09e-11	+0.12	Liver	SYPL2	1.23e-06	+0.50	Muscle_Skeletal	SARS	1.43e-04	+0.13

claude > thinking

The strongest row is CELSR2 in skeletal muscle, and it is not noise: that is a real signal for resting metabolic rate, via a muscle enhancer, different variant. Wrong trait here. LDL is set in the liver, so the tissue is fixed before I read a p value. But three liver genes' expressions rise with similar effect size, SORT1 +1.27, PSRC1 +1.24, CELSR2 +1.04, too close to call.

Occupancy can test whether the locus is regulatory at all, but it will not separate three promoters either.

Claude and ToolUniverse, turn by turn

```
claude > tool_use UCSC_get_tf_binding_clusters
{"genome": "hg38", "chrom": "chr1", "start": 109274768, "end": 109275168}
GENETICS · CALL 3 OF 4

tooluniverse > result (itemsReturned 118) Columns: factor, cluster score, number of ENCODE experiments
FOXA1      1000    7 exp      ZNF687     994    1 exp      XRCC5      695    1 exp      ZBTB7B     579    1 exp      HNF4G      454    1 exp
FOXA2      1000    4 exp      TBL1XR1    979    1 exp      RCOR1      647    2 exp      GATAD2A    558    1 exp      USF1       433    1 exp
CEBPB     1000    1 exp      PCBP1      964    1 exp      MBD2       632    1 exp      NBN        553    1 exp      ATF3       416    1 exp
ZFX        1000    1 exp      RBF0X2     875    2 exp      MNT        621    1 exp      SKIL       553    1 exp      SMARCE1    415    2 exp
RBM39      1000    1 exp      EP300      875    3 exp      KDM1A      614    1 exp      POLR2G     549    2 exp      NCOR1      414    2 exp
ZBTB7A     1000    2 exp      ZHX2       865    2 exp      TFAP4      611    1 exp      SIN3A      543    3 exp      ZNF512B    396    1 exp
DPF2       1000    3 exp      HDAC1      828    2 exp      TCF12      610    1 exp      LEF1       522    1 exp      SOX13      390    1 exp
MYC        1000    4 exp      CCAR2      776    1 exp      NR3C1      608    3 exp      MITF       510    2 exp      MTA3       388    1 exp
SP1        1000    4 exp      TCF7L2     753    1 exp      JUND       606    3 exp      LCORL      507    1 exp      RXRA       386    2 exp
HDAC2      1000    1 exp      POLR2A     736    13 exp   MTA2       602    1 exp      ASH2L      506    1 exp      SREBF1     381    1 exp
GATAD2B    1000    2 exp      CTBP1      720    2 exp      TCF7       592    1 exp      NEUROD1    493    1 exp      HNF4A      376    2 exp
HNRNPL     1000    1 exp      AGO2       706    1 exp      ZNF217     581    1 exp      MAX        485    2 exp      CTCF       143    4 exp

...and 58 further factors below score 376. Liver factors present: FOXA1, FOXA2, HNF4A, HNF4G, CEBPB, SREBF1, NR3C1.

claude > thinking
The liver programme is here, which is what makes a liver eQTL credible rather than incidental. But 112 of the 118 clusters contain the variant base, and CEBPB's score of 1000 comes from a single experiment whose cell background the track does not report – at 22% minor allele frequency, that background is not a detail. No track here carries allele information, so the call is consistent with the T allele creating a C/EBP site without testing it.
```

Claude and ToolUniverse, turn by turn

claude > tool_use PubMed_search_articles

GENETICS · CALL 4 OF 4

```
{"query": "rs12740374", "max_results": 40}
```

tooluniverse > result 14 records · then get_article_metadata on the three that bear on causality

PMID	YEAR	FIRST AUTHOR	JOURNAL	get_article_metadata 29097363 · abstract
20686566	2010	Musunuru K	Nature	
the T allele creates a C/EBP site and shifts hepatic SORT1 expression				"In primary human hepatocytes from multiple individuals heterozygous at rs12740374, we used CRISPR-Cas9 to specifically target the rs12740374 minor allele sequence ex vivo, resulting in a reproducible reduction in [SORT1] expression."
29097363	2017	Wang X	Arterioscler Thromb	
CRISPR ablation of the minor allele in heterozygous primary human hepatocytes				"We generated a locus-humanized transgenic mouse with a BAC bearing the human locus ... resulting in a substantial reduction of hepatic expression."
39018443	2025	Bandesh K	J Clin Endocrinol	
same locus in muscle, resting metabolic rate				

claude > thinking

The three database calls narrowed it to three genes and then stopped, because every one of them measures correlation across a locus. Only this result perturbs the variant and watches a single gene respond.

claude > answer

SORT1, in hepatocytes. CELSR2 is only where the variant sits, PSRC1 is statistically tied in liver, and neither survives ablation of the allele. The muscle CELSR2 signal is real but answers a different question.

A query from you?